香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	DDA3020
Course title (English)	Machine Learning
Course title (Chinese)	机器学习
Units	3
Language of Instruction	English
Description (English)	This is a fundamental course to provide the general concepts of machine learning. This course provides you the opportunity to learn skills and content to practice and engage in scalable machine learning methods on massive data and to study methods to train/learn/develop computational models from varieties of data. It covers supervised learning, unsupervised learning, decision trees, neural networks.
Description (Chinese)	本课程是教授机器学习概念之基础课程，透过修习本课程，可以学习机器学习的各种基本方法，并可以学习如何运用大量数据来训练机器学习模型。本课程主题包括监督学习、无监督学习、决策树、神经网络等。

2. Prerequisites / Co-requisites

A. Prerequisites

CSC1001 Introduction to Computer Science: Programming Methodology
CSC1002 Computational Laboratory
STA2001 Probability and Statistics I or STA2003 Probability

B. Co-requisites

None.

3. Learning Outcomes

Knowledge:

• Students will understand the key models and concepts of machine learning
• Students will understand the strengths and limitations of popular machine learning techniques
• Students will understand popular machine learning algorithms to solve the real-world problems
• Students will be able to identify promising applications of machine learning

Skills:
• Students will utilize a programming language to learn, visualize, and mine new insights from big data
• Students will utilize existing software to analyze available data to inform critical decisions
• Students will study data scientifically, and use it to prove hypotheses
• Students will be able to actively manage and participate in machine learning projects executed by consultants or specialists in machine learning

Valued/Attitude:
• Students will be equipped with some machine learning knowledge and be able to communicate with domain experts to solve their data science problems
• Students will be more vigilant in some data science issues and aware with the data science development in the society
• Students will have awareness of the impact of machine learning in social, industrial, environmental and technological context
• Students will be more literate in data science. They will develop a knowledge in data science so that they can disseminate data science related new development

4. Course syllabus

1.Review of probability and linear algebra: Gaussian models, models for discrete and continuous data, matrix multiplication/derivative
2.Supervised learning: linear regression and logistic regression, variable selection and LASSO, classification and regression trees, neural networks (MLP, CNN, RNN, and Transformer)
3.Unsupervised learning: clustering and mixture models, PCA and dimension reduction

5. Assessment Scheme

Component/ method	**% weight**
Assignments (Written homework)	30
Python/Scikit-learn Programming	30

Final exam	40

6. Grade descriptor

General

Grade	Description
A	Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

- A course questionnaire answered by every student after finishing all the lectures
- Reflections from teachers of more advanced courses
- Feedback from informal conversation over (free) coffee in my office and e-mail correspondence with students after class

8. Reading

Required Reading

1.Machine Learning: A Probabilistic Perspective, 0262018020, MIT, 2012, United States, https://probml.github.io/pml-book/

2.The Hundred-Page Machine Learning Book, 978-1777005474, HARD COVER ED., 2019, United States, http://themlbook.com/wiki/doku.php

9. Course components

Activity	Hours/week
Lectures	3
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
00000 00001	Introduction
00000 00002	Review: Probability, linear algebra, optimization
00000 00003	Linear regression
00000 00004	Logistic regression
00000 00005	Support vector machine
00000 00006	Decision tree and random forest
00000 00007	Neural networks I (MLP & CNN)
00000 00008	Neural networks II (RNN & Transformer)
00000 00009	Over-fitting, bias-variance trade-off
00000 00010	Performance evaluation
00000 00011	Introduction to unsupervised learning

00000 00012	K-means and Gaussian mixture models
00000 00013	Expectation Maximization
00000 00014	PCA

11. Any other information

12. Version date

Version number	6
As of (date)	2024/11/18